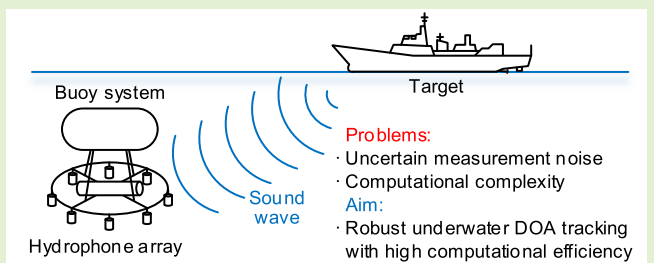


A Fast Variational Bayesian Adaptive Extended Kalman Filter for Robust Underwater Direction-of-Arrival Tracking

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Abstract—Underwater direction-of-arrival (DOA) tracking using hydrophone array is an important research subject in passive sonar signal processing. In this study, the robust underwater DOA tracking with the unknown measurement noise results from underwater environmental noise is concerned. To deal with the unknown measurement noise, the existing variational Bayesian adaptive extended Kalman filter (VB-AEKF) for robust underwater DOA tracking jointly estimates the covariance matrix of unknown measurement noise and the DOA by using variational Bayesian approach in the framework of extended Kalman filter. However, the iteration process of VB approach and the nonlinear measurement model based on hydrophone array signal result in significant computational complexity of the VB-AEKF for DOA tracking. To better solve the problem of robust DOA tracking with unknown measurement noise covariance matrix, a novel computationally efficient version of the existing VB-AEKF, which is named the fast variational Bayesian adaptive extended Kalman filter (FVB-AEKF), is proposed. An equivalent variational iteration process to estimate the measurement noise covariance matrix is derived. The proposed FVB-AEKF is identical to the VB-AEKF but reduces the computational complexity of the algorithm. The proposed method is tested via Monte Carlo simulations of an underwater DOA tracking scenario and an experiment in the South China Sea in July 2021. The results of the simulations and the experiment verified the accuracy and the computational efficiency of the FVB-AEKF.

Index Terms—Adaptive tracking, Kalman filter, nonlinear measurement model, underwater direction-of-arrival (DOA) tracking, variational Bayesian approach.



I. INTRODUCTION

DIRECTION-OF-ARRIVAL (DOA) estimation of underwater targets is an important research topic in passive sonar signal processing. For scenarios of a moving target, traditional DOA estimation methods cut the measurements of the received signal into sequential periods by comprehensively considering the change rate of the DOA and the number of snapshots to process them. However, the kinetic characteristics of the moving target are ignored [1], [2], [3], [4]. By utilizing

not only the measurements but also the motion model of the underwater target, DOA tracking methods can provide a more robust and accurate bearing angle estimation than traditional DOA estimation methods [5], [6], [7], [8], [9], [10], [11].

Considering the superior characteristics of DOA tracking methods, a number of research studies have been carried out. DOA tracking methods are mostly based on the Bayesian filter, and recursively estimate the state (bearing angle of the target) from the array-signal-based measurement (output signal of the hydrophone array containing the signal emitted by the target) in real time. To deal with the nonlinear relation between the bearing angle of the target and the measurement, i.e., nonlinear measurement model, many tracking techniques have been proposed based on nonlinear Bayesian filters, such as extended Kalman filter (EKF) [5], unscented Kalman filter (UKF) [12], cubature Kalman filter (CKF) [13], and particle filter (PF) [6]. Among these nonlinear Bayesian filters, the EKF is both robust and computationally efficient due to its simple algorithm principle. Thus, the EKF is widely used in engineering applications.

In most research studies of underwater tracking, the measurement noise is usually assumed to be subject to a zero-mean Gaussian distribution with a known fixed covariance matrix.

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However, this measurement noise covariance matrix (MNCM) may become unknown and unstable in real underwater tracking scenarios because of the unknown underwater environmental noise. The mismatch between the actual unknown and unstable MNCM and the assumed known fixed MNCM will significantly degrade the tracking performance. In order to deal with unknown and unstable MNCM, a number of adaptive Kalman filters based on different algorithms are proposed by estimating the MNCM. Zhang et al. [7] used the Sage-Husa (SH) algorithm [14] to estimate the MNCM and carried out a robust underwater DOA tracking method by substituting the estimated MNCM into EKF. By estimating the MNCM in a maximum likelihood (ML) framework, the expectation maximum adaptive Kalman filter (EM-AKF) was proposed [15], [16]. Sarkka and Nummenmaa [19], and Hartikainen [20] assumed the MNCM to be subject to an inverse Wishart distribution and iteratively estimated the MNCM by using a variational Bayesian approach [17], [18], namely the variational Bayesian adaptive Kalman filter (VB-AKF). Huang et al. [21] assumed the mean square error matrix to be also subject to an inverse Wishart distribution and jointly estimate it along with MNCM to improve the performance of VB-AKF. In the existing works, the superior accuracy and stability of the VB-AKF were demonstrated [19], [20], [21]. However, due to the iteration process of the variational Bayesian approach, the VB-AKF has significant computation for high-dimensional systems. In order to address this problem, Xu et al. [22] proposed a computationally efficient VB-AKF for a liner transfer alignment system with a high-dimensional state. In addition, based on the VB-AKF [19], Hou et al. [23] derived a variational Bayesian adaptive extended Kalman filter (VB-AEKF) to adaptively estimate the MNCM for robust underwater DOA tracking.

In this article, robust underwater DOA tracking with regard to unknown measurement noise was studied. The abovementioned VB-AEKF for DOA tracking [23] is also proposed for the same task. Similar to VB-AKF, the VB-AEKF for DOA tracking uses the variational Bayesian approach in the framework of the extended Kalman filter to jointly estimate the state and the covariance of measurement noise, and the MNCM needs to update by several variational iterations to ensure the estimation accuracy. It should be noted that the array-signal-based measurements bring a high-dimensional and complicated nonlinear measurement model, which leads to significant computations of the measurement function and the Jacobian matrix. For such a measurement model, the variational iteration process of the VB-AEKF results in a high computational overload. This problem goes against the application of VB-AEKF for DOA tracking.

To avoid this issue of VB-AEKF for DOA tracking, a fast VB-AEKF (FVB-AEKF) for DOA tracking is proposed in this article. Inspired by the computationally efficient VB-AKF [22] for linear systems with high-dimensional states, we derive a variational iteration process equivalent to that of the VB-AEKF but more computationally efficient for DOA tracking with a high-dimensional and nonlinear measurement model. The proposed variational iteration process degrades the number of computations by reducing the iterations of parameters, such as the measurement function and its Jacobian matrix. Based on this novel variational iteration process, the

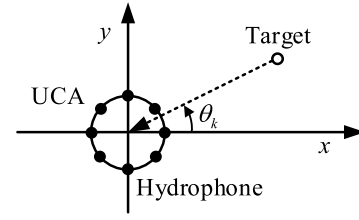


Fig. 1. Configuration of the UCA-based measurement system.

FVB-AEKF is proposed. Rich simulations and experiments are carried out in the underwater DOA tracking scenario with unknown measurement noise, and the superior performance of the proposed FVB-AEKF both in terms of robustness and computational efficiency is verified.

The rest of this article is organized as follows: In Section II, the underwater DOA tracking problem is formulated. In Section III, the VB-AEKF for robust underwater DOA tracking is derived in detail, and the number of computations of it is analyzed. The innovative fast method for robust underwater DOA tracking, namely the FVB-AEKF, is proposed in Section IV. In Sections V and VI, comprehensive simulations and an experiment in the South China Sea in July 2021 are conducted to verify the superior performance of FVB-AEKF in terms of accuracy and computational efficiency. Finally, in Section VII, the conclusions are drawn.

II. PROBLEM FORMULATION

A. Kinematic Model of the Underwater Target by the Bearing Angle

The motion process of the underwater target is considered to be a stochastic process. The bearing angle of the target is assumed to perform in the constant velocity (CV) motion mode.

Let θ_k denotes the bearing angle of the target and $\dot{\theta}_k$ denotes the change rate of θ_k at time step k , the system state is expressed as $\mathbf{x}_k = (\theta_k, \dot{\theta}_k)^T$. Then, the CV motion kinematic model of the bearing angle is expressed as

$$\mathbf{x}_k = \mathbf{F}_{k|k-1}\mathbf{x}_{k-1} + \mathbf{G}_k\mathbf{w}_k \quad (1)$$

where $\mathbf{F}_{k|k-1}$ is the state transition matrix, $\mathbf{x}_{k-1}$ is the system state at time step $k-1$, $\mathbf{G}_k$ is the noise driving matrix, and $\mathbf{w}_k$ denotes the zero mean Gaussian process noise with the covariance matrix $\mathbf{Q}_k$ caused by the unknown underwater environment. $\mathbf{Q}_k$ is usually set by engineering experience in target tracking scenario. $\mathbf{F}_{k|k-1}$ and $\mathbf{G}_k$ is given by

$$\mathbf{F}_{k|k-1} = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} \quad (2)$$

$$\mathbf{G}_k = \begin{bmatrix} T^2/2 \\ T \end{bmatrix} \quad (3)$$

where T is the interval between the adjacent time steps.

B. Measurement Model Based on the Received Signal of the UCA

Considering the bearing angle of the target varies in a wide range in a long-time tracking scenario, the uniform circular array (UCA) is used since it is free from the port and starboard ambiguity problem and possesses uniform aperture at all bearing angles [24], [25].

The configuration of the UCA and the underwater target is shown in Fig. 1. Considering the target emits a narrowband acoustic signal at the frequency f with the amplitude a and the initial phase ϕ_0 , the target signal at k is expressed as

$$s(k) = \alpha \exp(j(2\pi f k T + \phi_0)). \quad (4)$$

The emitted signal given by (4) is received by the UCA with P elements at time step k . The P elements are uniformly placed around the circle with the radius R and the center at the origin of the cartesian coordinates. It is assumed that the range of the target satisfies the far-field condition to regard the arrived signal as a plane wave, and the sound velocity is c .

Then, the measurement vector of the received signal at time step k can be expressed as

$$\mathbf{z}_k = \mathbf{h}_k(\mathbf{x}_k) + \mathbf{v}_k = \text{real}(\mathbf{A}(\theta_k)s(k)) + \mathbf{v}_k \quad (5)$$

where $\mathbf{h}_k(\mathbf{x}_k)$ denotes the nonlinear measurement function, $\mathbf{A}(\theta_k)$ is the array manifold vector of the UCA, and $\mathbf{v}_k$ denotes the measurement noise. $\mathbf{A}(\theta_k)$ is given by

$$\begin{cases} \mathbf{A}(\theta_k) = [A_1(\theta_k), A_2(\theta_k), \dots, A_P(\theta_k)]^T \\ A_p(\theta_k) = \exp(j2\pi f r/c \cos(\theta_k - 2\pi p/P)) \end{cases} \quad (6)$$

where θ_k is the angle between the target and the positive x -axis with the positive counterclockwise direction, p is the number of elements and $p = 1, 2, \dots, P$.

The measurement noise $\mathbf{v}_k$ at the time step k is defined as a zero mean Gaussian noise with a P -order covariance matrix $\mathbf{R}_k$. The MNM $\mathbf{R}_k$ is usually assumed to be fixed value by engineering experience in most tracking problems. However, in the real underwater tracking scenario, the unknown underwater environment can result in the fluctuant MNM, which is hard to deal with by using traditional tracking techniques. This problem seriously degrades the performance of traditional tracking techniques.

III. COMPUTATIONAL LOAD ANALYSIS VB-AEKF FOR DOA TRACKING

By using the VB iteration approach, the existing VB-AEKF jointly estimates the state and the MNM, whose probability distributions are assumed to be Gaussian distribution and inverse Wishart distribution, respectively. According to [23], the pseudo-code of the VB-AEKF for DOA tracking using UCA at a one time step is given in Algorithm 1.

The meaning of input parameters is given as follows: $\hat{\mathbf{x}}_{k-1|k-1}$ is the state of the target at the time step $k-1$, $\mathbf{P}_{k-1|k-1}$ is the mean-square error matrix at the time step $k-1$, $\mathbf{z}_k$ is the measurement vector at time step k , $\mathbf{Q}_k$ is the process noise covariance matrix. $\hat{\mathbf{u}}_{k-1|k-1}$, $\hat{\mathbf{U}}_{k-1|k-1}$, ρ and N are the parameters for the variational Bayesian approach to estimate the MNM at a time step $k-1$. $\hat{\mathbf{u}}_{k-1|k-1}$ and $\hat{\mathbf{U}}_{k-1|k-1}$ is the degree of free (dof) and the prior inverse scale matrix for the variational Bayesian approach, respectively. ρ is the forgetting factor ($\rho \in (0, 1]$), and N is the number of VB iterations.

The meaning of output parameters is given as follows: $\hat{\mathbf{x}}_{k|k}$ is the state of the target at time step k , $\mathbf{P}_{k|k}$ is the mean-square error matrix at time step k . $\hat{\mathbf{u}}_{k|k}$ and $\hat{\mathbf{U}}_{k|k}$ are the updated dof and the prior inverse scale matrix.

Algorithm 1 One Time Step of the VB-AEKF for DOA Tracking Using UCA

Inputs: $\hat{\mathbf{x}}_{k-1|k-1}$, $\mathbf{P}_{k-1|k-1}$, $\mathbf{z}_k$, $\hat{\mathbf{u}}_{k-1|k-1}$, $\hat{\mathbf{U}}_{k-1|k-1}$, $\mathbf{Q}_k$, ρ , N .
Time update

1. $\hat{\mathbf{x}}_{k-1|k-1} = \mathbf{F}_{k-1|k-1} \hat{\mathbf{x}}_{k-1|k-1}$.
2. $\mathbf{P}_{k|k-1} = \mathbf{F}_{k|k-1} \mathbf{P}_{k-1|k-1} \mathbf{F}_{k|k-1}^T + \mathbf{G}_k \mathbf{Q}_k \mathbf{G}_k^T$.
3. $\hat{\mathbf{u}}_{k-1|k-1} = \rho(\hat{\mathbf{U}}_{k-1|k-1} - n - 1) + n + 1$, $\hat{\mathbf{U}}_{k-1|k-1} = \rho \hat{\mathbf{U}}_{k-1|k-1}$.

Iteration measurement update

4. Initialization: $\hat{\mathbf{x}}_{k|k}^{(0)} = \hat{\mathbf{x}}_{k|k-1}$, $\hat{\mathbf{R}}_k^{(0)} = \hat{\mathbf{U}}_{k|k-1} / \hat{\mathbf{u}}_{k|k-1}$.

For $i = 0 : N - 1$

Update $q^{(i+1)}(\mathbf{x}_k) = \text{Nl}(\mathbf{x}_k; \mathbf{x}_{k|k}^{(i+1)}, \mathbf{P}_{k|k}^{(i+1)})$

5. Calculate $\hat{\mathbf{s}}_{k|k}^{(i)}(k)$, $\mathbf{h}(\hat{\mathbf{x}}_{k|k}^{(i)})$, and $\mathbf{H}_k^{(i)}$.
6. $\mathbf{K}_k^{(i+1)} = \mathbf{P}_{k|k-1} (\mathbf{H}_k^{(i)})^T (\mathbf{H}_k^{(i)} \mathbf{P}_{k|k-1} (\mathbf{H}_k^{(i)})^T + \hat{\mathbf{R}}_k^{(i)})^{-1}$,
7. $\hat{\mathbf{x}}_{k|k}^{(i+1)} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k^{(i+1)} (\mathbf{z}_k - \mathbf{h}(\hat{\mathbf{x}}_{k|k}^{(i)}) - \mathbf{H}_k^{(i)} (\hat{\mathbf{x}}_{k|k-1} - \hat{\mathbf{x}}_{k|k}^{(i)}))$,
8. $\mathbf{P}_{k|k}^{(i+1)} = \mathbf{P}_{k|k-1} - \mathbf{K}_k^{(i+1)} \mathbf{H}_k^{(i)} \mathbf{P}_{k|k-1}$.

Update $q^{(i+1)}(\mathbf{R}_k) = \text{IW}(\mathbf{R}_k; \hat{\mathbf{u}}_{k|k}^{(i+1)}, \hat{\mathbf{U}}_{k|k}^{(i+1)})$

9. Calculate $\hat{\mathbf{s}}_{k|k}^{(i+1)}(k)$, $\mathbf{h}(\hat{\mathbf{x}}_{k|k}^{(i+1)})$, and $\mathbf{H}_k^{(i+1)}$.
10. $\mathbf{B}_k^{(i+1)} = (\mathbf{z}_k - \mathbf{h}(\hat{\mathbf{x}}_{k|k}^{(i+1)})^T \mathbf{H}_k^{(i+1)} \mathbf{P}_{k|k}^{(i+1)} (\mathbf{H}_k^{(i+1)})^T)$,
11. $\hat{\mathbf{U}}_{k|k}^{(i+1)} = \hat{\mathbf{U}}_{k|k} + \mathbf{B}_k^{(i+1)}$, $\hat{\mathbf{u}}_{k|k}^{(i+1)} = \hat{\mathbf{u}}_{k|k-1} + 1$,
12. $\hat{\mathbf{R}}_k^{(i+1)} = \hat{\mathbf{U}}_{k|k}^{(i+1)} / \hat{\mathbf{u}}_{k|k}^{(i+1)}$.

End for

13. $\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k}^{(N)}$, $\mathbf{P}_{k|k} = \mathbf{P}_{k|k}^{(N)}$, $\hat{\mathbf{u}}_{k|k} = \hat{\mathbf{u}}_{k|k}^{(N)}$, $\hat{\mathbf{U}}_{k|k} = \hat{\mathbf{U}}_{k|k}^{(N)}$.

Outputs: $\hat{\mathbf{x}}_{k|k}$, $\mathbf{P}_{k|k}$, $\hat{\mathbf{u}}_{k|k}$, $\hat{\mathbf{U}}_{k|k}$.

TABLE I
FLOPS OF COMMON MATRIX CALCULATIONS

Matrices	Calculation	FLOP _A	FLOP _M	FLOP _{SR}
$\mathbf{A} \in \mathbf{R}^{P \times P}$, $\mathbf{B} \in \mathbf{R}^{P \times P}$	$\mathbf{A} \pm \mathbf{B}$	$\beta\gamma$	/	/
$\mathbf{A} \in \mathbf{R}^{P \times P}$, $\mathbf{B} \in \mathbf{R}^{P \times P}$	$\mathbf{AB}$	$\beta\eta(\gamma-1)$	$\beta\gamma\eta$	/
$\mathbf{A} \in \mathbf{R}^{P \times P}$	$\mathbf{A}^{-1}$	$0.5\beta^3 - 0.5\beta^2$	$0.5\beta^3 + 1.5\beta^2$	β

In Algorithm 1, $\text{Nl}(\cdot; \mu, \Sigma)$ denotes the probability distribution function (pdf) of the Gaussian distribution with mean μ and covariance matrix Σ , $\text{IW}(\cdot; \lambda, \Psi)$ denotes the pdf of the inverse Wishart distribution with dof λ and inverse scale matrix Ψ (details of the pdf of the IW distribution could be found in [21]), forgetting factor ρ is to spread the previous posterior parameters $\hat{\mathbf{u}}_{k-1|k-1}$ and $\hat{\mathbf{U}}_{k-1|k-1}$ to prior parameters $\hat{\mathbf{u}}_{k-1|k-1}$ and $\hat{\mathbf{U}}_{k-1|k-1}$, which indicates the extent of the time-fluctuations of the MNM. The expression of $\hat{\mathbf{s}}_{k|k}^{(i)}(k)$, $\mathbf{h}(\hat{\mathbf{x}}_{k|k}^{(i)})$ and $\mathbf{H}_k^{(i)}$ are given in the Appendix. Moreover, step 5 of the algorithm is obviously unnecessary to operate after the first iteration because $\hat{\mathbf{s}}_{k|k}^{(i)}(k)$, $\mathbf{h}(\hat{\mathbf{x}}_{k|k}^{(i)})$ and $\mathbf{H}_k^{(i)}$ have already been calculated in the last iteration.

In computer calculation, the total number of floating-point operations (FLOPs), such as addition, multiplication, square root, and trigonometric function, can represent the computational complexity of the algorithm [25], [26]. The FLOPs of common matrix calculations are given in Table I, where

TABLE II
FLOPS OF THE VB-AEKF FOR DOA TRACKING

	Parameters	FLOP _A	FLOP _M	FLOP _D	FLOP _T
Time update	$\hat{\mathbf{x}}_{k k-1}$	$a^2 - a$	a^2	/	/
	$\mathbf{P}_{k k-1}$	$2a^3 - a^2$	$2a^3 + a^2 + a$	/	/
	$\hat{\mathbf{u}}_{k k-1}$	/	1	/	/
	$\hat{\mathbf{U}}_{k k-1}$	/	b^2	/	/
Initialization	$\hat{\mathbf{R}}_k^{(0)}$	/	/	b^2	/
Measurement function and Jacobian matrix	$\hat{\mathbf{s}}_{k k}^{(i)}(k)$	$10b - 1$	$12b + 6$	2	/
	$\mathbf{h}(\hat{\mathbf{x}}_{k k}^{(i)})$	$3b$	$5b$	/	$3b$
	$\mathbf{H}_k^{(i)}$	$3b$	$7b$	/	$4b$
Iteration process	$\mathbf{K}_k^{(i+1)1}$	$2a^2b + 2ab^2 - 3ab + 0.5b^3 - 0.5b^2$	$2a^2b + 2ab^2 + 0.5b^3 + 1.5b^2$	/	/
	$\hat{\mathbf{x}}_{k k}^{(i+1)}$	$ab + b$	ab	/	/
	$\mathbf{P}_{k k}^{(i+1)}$	$a^3 + a^2b - a^2$	$a^3 + a^2b$	/	/
	$\mathbf{B}_k^{(i+1)}$	$a^2b + ab^2 - ab + b$	$a^2b + ab^2 + b^2$	/	/
	$\hat{\mathbf{U}}_{k k}^{(i+1)}$	b^2	/	/	/
	$\hat{\mathbf{R}}_k^{(i+1)}$	/	/	b^2	/

¹ $\mathbf{K}_k^{(i+1)}$ costs b times of FLOP_{SR} besides the FLOPs given in the table.

FLOP_A, FLOP_M, and FLOP_{SR} denote the FLOPs of addition, multiplication, and square-root operation, respectively, and the inverse of the matrix is calculated using the Cholesky decomposition.

According to Algorithm 1 and Table I, the number of FLOPs of the VB-AEKF for DOA tracking at a one time step is derived. Significantly, $\hat{\mathbf{s}}_{k|k}^{(i)}(k)$, $\mathbf{h}(\hat{\mathbf{x}}_{k-1|k-1})$ and $\mathbf{H}_k^{(i)}$ involve complex calculations. The addition of complex numbers costs 2 FLOP_{AS}, the multiplication of complex numbers costs 2 FLOP_{AS} and 4 FLOP_{MS}, the inversion of a complex number costs a FLOP_A, 2 FLOP_{MS} and 2 FLOP_{DS} (FLOPs of division), and the exponent of a complex number costs 2 FLOP_{TS} (FLOPs of trigonometric function). Then, the FLOPs of the VB-AEKF for DOA tracking at a one time step are given in Table II, where a and b denote the dimension of the state vector $\mathbf{x}_k$ and the measurement vector $\mathbf{z}_k$, respectively. $2\pi fr/c$ and $2\pi p/P$ in the calculation of $\mathbf{h}(\hat{\mathbf{x}}_{k-1|k-1})$ and $\mathbf{H}_k^{(i)}$ can both be calculated just one time at the beginning of the algorithm, which carried out insignificant computational load and thus the computations of them are ignored.

According to Table II, the total number of computations of the VB-AEKF at a one time step are given as follows:

$$\begin{cases}
 \text{FLOP}_A = N(a^3 + 4a^2b + 3ab^2 + 0.5b^3 - 3ab + 0.5b^2 + 2b) \\
 \quad + (N+1)(16b - 1) + 2a^3 - a \\
 \text{FLOP}_M = N(a^3 + 4a^2b + 3ab^2 + 0.5b^3 + ab + 2.5b^2) \\
 \quad + (N+1)(24b + 6) + 2a^3 + 2a^2 + b^2 + a + 1 \\
 \text{FLOP}_D = b^2 + 2(N+1) + Nb^2 \\
 \text{FLOP}_{SR} = Nb \\
 \text{FLOP}_T = 7(N+1)b.
 \end{cases} \quad (7)$$

As can be seen from Algorithm 1, the high-dimensional measurement function $\mathbf{h}(\hat{\mathbf{x}}_{k|k}^{(i)})$ and Jacobian matrix $\mathbf{H}_k^{(i)}$ need to be calculated to obtain $\mathbf{P}_{k|k}^{(i+1)}$ for the parameter $\mathbf{B}_k^{(i+1)}$ in

every variational iteration. Besides, the trigonometric functions in the calculations of $\mathbf{h}(\hat{\mathbf{x}}_{k|k}^{(i)})$ and $\mathbf{H}_k^{(i)}$ bring much more computational load than additions, and even more than multiplications and square roots. Thus, the trigonometric function operations and the high-dimensional measurement function and the Jacobian matrix result in high computational complexity. In order to solve this problem, we derive a novel iteration process in Section IV to avoid the calculation of $\mathbf{P}_{k|k}^{(i+1)}$ and reduce the iterations of $\mathbf{h}(\hat{\mathbf{x}}_{k|k}^{(i)})$ and $\mathbf{H}_k^{(i)}$.

IV. FAST VB-AEKF FOR DOA TRACKING

A. Formula Derivation

The above analyses proved that the iterations of $\mathbf{P}_{k|k}^{(i+1)}$ and the calculations of $\mathbf{h}(\hat{\mathbf{x}}_{k|k}^{(i)})$ and $\mathbf{H}_k^{(i)}$ result in the high computational complexity of the VB-AEKF for DOA tracking given in Algorithm 1. Similar to the main thought of computationally efficient VB-AKF [22], we derived a fast variational iteration process equivalent to VB-AEKF. Different from the computationally efficient VB-AKF for linear systems with high-dimensional states, this novel fast variational iteration process is suitable for DOA tracking with nonlinear and high-dimensional measurement. The novel fast variational iteration process improves the computational efficiency by calculating $\mathbf{B}_k^{(i+1)}$ for the MNCM estimation without iteratively calculating $\mathbf{P}_{k|k}^{(i+1)}$ and reducing the iterations of $\mathbf{h}(\hat{\mathbf{x}}_{k|k}^{(i)})$ and $\mathbf{H}_k^{(i)}$. Then, an FVB-AEKF for DOA tracking (FVB-AEKF) is proposed by using this novel iteration process. The derivation of the novel iteration process is given in detail as follows.

According to the equation for calculation of $\mathbf{B}_k^{(i+1)}$ in step 14 of Algorithm 1, we respectively derive the two terms on the right side of the equation $(\mathbf{z}_k - \mathbf{h}(\hat{\mathbf{x}}_{k|k}^{(i+1)}))(\mathbf{z}_k - \mathbf{h}(\hat{\mathbf{x}}_{k|k}^{(i+1)}))^T$ and $\mathbf{H}_k^{(i+1)} \mathbf{P}_{k|k}^{(i+1)} (\mathbf{H}_k^{(i+1)})^T$ as follows.

The first-order Taylor expansion of $\mathbf{h}(\hat{\mathbf{x}}_{k|k}^{(i+1)})$ at $\hat{\mathbf{x}}_{k|k}$ is

$$\mathbf{h}(\hat{\mathbf{x}}_{k|k}^{(i+1)}) = \mathbf{h}(\hat{\mathbf{x}}_{k|k-1}) + \mathbf{H}_{k|k-1}(\hat{\mathbf{x}}_{k|k}^{(i+1)} - \hat{\mathbf{x}}_{k|k-1}). \quad (8)$$

Moreover, according to the standard EKF, the measurement update of the state $\hat{\mathbf{x}}_{k|k}^{(i+1)}$ is expressed as

$$\hat{\mathbf{x}}_{k|k}^{(i+1)} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k^{(i+1)} (\mathbf{z}_k - h(\hat{\mathbf{x}}_{k|k-1})). \quad (9)$$

Substituting (8) and (9) into $\mathbf{z}_k - h(\hat{\mathbf{x}}_{k|k}^{(i+1)})$, we obtained

$$\begin{aligned} \mathbf{z}_k - h(\hat{\mathbf{x}}_{k|k}^{(i+1)}) &= \mathbf{z}_k - h(\hat{\mathbf{x}}_{k|k}) - \mathbf{H}_{k|k-1} (\hat{\mathbf{x}}_{k|k}^{(i+1)} - \hat{\mathbf{x}}_{k|k-1}) \\ &= \mathbf{z}_k - h(\hat{\mathbf{x}}_{k|k-1}) - \mathbf{H}_{k|k-1} \mathbf{K}_k^{(i+1)} (\mathbf{z}_k - h(\hat{\mathbf{x}}_{k|k-1})) \\ &= (\mathbf{I}_b - \mathbf{H}_{k|k-1} \mathbf{K}_k^{(i+1)}) (\mathbf{z}_k - h(\hat{\mathbf{x}}_{k|k-1})). \end{aligned} \quad (10)$$

Then, the term $(\mathbf{z}_k - h(\hat{\mathbf{x}}_{k|k}^{(i+1)}))(\mathbf{z}_k - h(\hat{\mathbf{x}}_{k|k}^{(i+1)}))^T$ in step 14 of Algorithm 1 is expressed as

$$\begin{aligned} &(\mathbf{z}_k - h(\hat{\mathbf{x}}_{k|k}^{(i+1)}))(\mathbf{z}_k - h(\hat{\mathbf{x}}_{k|k}^{(i+1)}))^T \\ &= (\mathbf{I}_b - \mathbf{H}_{k|k-1} \mathbf{K}_k^{(i+1)}) \tilde{\mathbf{P}}_{k|k-1}^{zz} (\mathbf{I}_b - \mathbf{H}_{k|k-1} \mathbf{K}_k^{(i+1)})^T \end{aligned} \quad (11)$$

where

$$\tilde{\mathbf{P}}_{k|k-1}^{zz} = (\mathbf{z}_k - h(\hat{\mathbf{x}}_{k|k-1}))(\mathbf{z}_k - h(\hat{\mathbf{x}}_{k|k-1}))^T. \quad (12)$$

In order to derive the expression of the term $\mathbf{H}_k^{(i+1)} \mathbf{P}_{k|k}^{(i+1)} (\mathbf{H}_k^{(i+1)})^T$ in step 14 of Algorithm 1, we first give the mean square error matrix $\tilde{\mathbf{P}}_{k|k}^{zz}$ and the Kalman filter gain $\mathbf{K}_k^{(i+1)}$ according to the standard EKF as follows:

$$\tilde{\mathbf{P}}_{k|k}^{(i+1)} = \mathbf{P}_{k|k-1} - \mathbf{K}_k^{(i+1)} \mathbf{H}_k^{(i)} \mathbf{P}_{k|k-1} \quad (13)$$

$$\begin{aligned} \mathbf{K}_k^{(i+1)} &= \mathbf{P}_{k|k-1} (\mathbf{H}_k^{(i)})^T (\mathbf{H}_k^{(i)} \mathbf{P}_{k|k-1} (\mathbf{H}_k^{(i)})^T + \hat{\mathbf{R}}_k^{(i)})^{-1} \\ &\approx \mathbf{P}_{k|k-1} \mathbf{H}_{k|k-1}^T (\mathbf{H}_k^{(i)} \mathbf{P}_{k|k-1} (\mathbf{H}_k^{(i)})^T + \hat{\mathbf{R}}_k^{(i)})^{-1}. \end{aligned} \quad (14)$$

Then, the term $\mathbf{H}_{k|k}^{(i+1)} \hat{\mathbf{P}}_{k|k}^{(i+1)} (\mathbf{H}_{k|k}^{(i+1)})^T$ is derived as follows:

$$\begin{aligned} &\mathbf{H}_{k|k}^{(i+1)} \hat{\mathbf{P}}_{k|k}^{(i+1)} (\mathbf{H}_{k|k}^{(i+1)})^T \\ &\approx \mathbf{H}_{k|k-1} \hat{\mathbf{P}}_{k|k}^{(i+1)} \mathbf{H}_{k|k-1}^T \\ &= \mathbf{H}_{k|k-1} \mathbf{P}_{k|k-1} \mathbf{H}_{k|k-1}^T - \mathbf{H}_{k|k-1} \mathbf{K}_k^{(i+1)} \mathbf{H}_{k|k-1} \mathbf{P}_{k|k-1} \mathbf{H}_{k|k-1}^T \\ &= \mathbf{H}_{k|k-1} \mathbf{P}_{k|k-1} \mathbf{H}_{k|k-1}^T \\ &\quad - \mathbf{H}_{k|k-1} \mathbf{P}_{k|k-1} \mathbf{H}_{k|k-1}^T (\mathbf{P}_{k|k-1}^{zz(i+1)})^{-1} \mathbf{H}_{k|k-1} \mathbf{P}_{k|k-1} \mathbf{H}_{k|k-1}^T \end{aligned} \quad (15)$$

where

$$\mathbf{P}_{k|k-1}^{zz(i+1)} = \mathbf{H}_k^{(i)} \mathbf{P}_{k|k-1} (\mathbf{H}_k^{(i)})^T + \hat{\mathbf{R}}_k^{(i)}. \quad (16)$$

Let

$$\mathbf{D}_k^{(i)} = \mathbf{H}_k^{(i)} \mathbf{P}_{k|k-1} (\mathbf{H}_k^{(i)})^T - \mathbf{H}_{k|k-1} \mathbf{P}_{k|k-1} \mathbf{H}_{k|k-1}^T \quad (17)$$

$$\tilde{\mathbf{R}}_k^{(i)} = \mathbf{D}_k^{(i)} + \hat{\mathbf{R}}_k^{(i)} \quad (18)$$

then $\mathbf{P}_{k|k-1}^{zz(i+1)}$ given by (16) can be expressed as

$$\begin{aligned} &\mathbf{P}_{k|k-1}^{zz(i+1)} \\ &= \mathbf{H}_{k|k-1} \mathbf{P}_{k|k-1} \mathbf{H}_{k|k-1}^T - \mathbf{H}_{k|k-1} \mathbf{P}_{k|k-1} \mathbf{H}_{k|k-1}^T \\ &\quad + \mathbf{H}_k^{(i)} \mathbf{P}_{k|k-1} (\mathbf{H}_k^{(i)})^T + \hat{\mathbf{R}}_k^{(i)} \\ &= \mathbf{H}_{k|k-1} \mathbf{P}_{k|k-1} \mathbf{H}_{k|k-1}^T + \mathbf{D}_k^{(i)} + \hat{\mathbf{R}}_k^{(i)} \\ &= \mathbf{H}_{k|k-1} \mathbf{P}_{k|k-1} \mathbf{H}_{k|k-1}^T + \tilde{\mathbf{R}}_k^{(i)}. \end{aligned} \quad (19)$$

Similar to the derivation given by (13)–(20) of [22], we obtain

$$\mathbf{B}_k^{(i+1)} = \tilde{\mathbf{R}}_k^{(i)} + \mathbf{C}_k^{(i+1)} (\tilde{\mathbf{P}}_{k|k-1}^{zz} - \mathbf{P}_{k|k-1}^{zz(i+1)}) (\mathbf{C}_k^{(i+1)})^T \quad (20)$$

where

$$\mathbf{C}_k^{(i+1)} = \tilde{\mathbf{R}}_k^{(i)} (\mathbf{P}_{k|k-1}^{zz(i+1)})^{-1}. \quad (21)$$

Let

$$\Delta \mathbf{R}_k^{(i+1)} = \mathbf{C}_k^{(i+1)} (\tilde{\mathbf{P}}_{k|k-1}^{zz} - \mathbf{P}_{k|k-1}^{zz(i+1)}) (\mathbf{C}_k^{(i+1)})^T \quad (22)$$

then (20) can be expressed as

$$\mathbf{B}_k^{(i+1)} = \tilde{\mathbf{R}}_k^{(i)} + \Delta \mathbf{R}_k^{(i+1)}. \quad (23)$$

Substituting (23) into Algorithm 1, the algorithm of the FVB-AEKF is obtained. In order to reduce the calculation complexity further, we stop the update of $\mathbf{H}_k^{(i)}$ when the deviation between $\mathbf{H}_k^{(i-1)}$ and $\mathbf{H}_k^{(i)}$ is less than a threshold, i.e.,

$$\varepsilon = \|\mathbf{H}_k^{(i)} - \mathbf{H}_k^{(i-1)}\|_2 < \delta \quad (24)$$

where $\|\cdot\|_2$ denotes the two-norm of the matrix, δ denotes the threshold.

According to the derivations above, the pseudo-code of the FVB-AEKF for DOA tracking using UCA at one time step is given in Algorithm 2.

B. Computational Load Analysis of the VB-AEKF

According to Algorithm 2, the number of computations of all calculation steps in one time step of the FVB-AEKF for DOA tracking is derived in Table III. Thus, the total number of computations of the FVB-AEKF for DOA tracking at one time step is given as follows:

$$\left\{ \begin{aligned} \text{FLOP}_A &= N(2a^2b + 2ab^2 + 3.5b^3 - 2ab - 2.5b^2 + 14b - 1) \\ &\quad + N_H(2a^2b + 2ab^2 - 2ab - b^2) + 3a^3 + 3a^2b \\ &\quad + 2ab^2 + 0.5b^3 - a^2 - 2ab - 0.5b^2 - a + 5b \\ \text{FLOP}_M &= N(2a^2b + 2ab^2 + 3.5b^3 + 2.5b^2 + 19b + 6) \\ &\quad + N_H(2a^2b + 2ab^2) + 3a^3 + 3a^2b + 2ab^2 + 0.5b^3 \\ &\quad + 2a^2 + ab + 4.5b^2 + a + 5b + 1 \\ \text{FLOP}_D &= b^2 + N(2 + b^2) \\ \text{FLOP}_{SR} &= (N + 1)b \\ \text{FLOP}_T &= 3b + 4N_Hb \end{aligned} \right. \quad (25)$$

where N_H denotes the iterations number of $\mathbf{H}_k^{(i)}$. Setting the element number and the iteration number to 12 and 10, respectively, we have $a = 2$, $b = 12$, and $N_H \leq N = 10$. By substituting them into (7) and (25), the total number of computations of the VB-AEKF and the FVB-AEKF for DOA tracking are compared in Table IV.

Table IV shows that although the number of FLOP_{As} and FLOP_{Ms} of the FVB-AEKF is greater than that of the VB-AEKF, the number of FLOP_{Ts} of the FVB-AEKF is less than that of the VB-AEKF, while the number of FLOP_{Ds} and FLOP_{SRs} of VB-AEKF and FVB-AEKF are almost the same. Since computers utilize the Taylor expansion to operate

TABLE III
FLOPS OF THE VB-AEKF FOR DOA TRACKING

	Parameters	FLOP _A	FLOP _M	FLOP _D	FLOP _T
Time update	$\hat{\mathbf{x}}_{k k-1}$	$a^2 - a$	a^2	/	/
	$\mathbf{P}_{k k-1}$	$2a^3 - a^2$	$2a^3 + a^2 + a$	/	/
	$\hat{u}_{k k-1}$	/	1	/	/
	$\hat{\mathbf{U}}_{k k-1}$	/	b^2	/	/
Initialization	$\hat{\mathbf{R}}_k^{(0)}$	/	/	b^2	/
Measurement function and Jacobian matrix	$\hat{\mathbf{P}}_{k k-1}^{zz}$	b	b^2	/	/
	$\hat{s}_{k k-1}(k)$ or $\hat{s}_{k k}^{(i)}(k)$	$10b - 1$	$12b + 6$	2	/
	$\mathbf{h}(\hat{\mathbf{x}}_{k k-1})$	$3b$	$5b$	/	$3b$
	$\mathbf{H}_{k k-1}$ or $\mathbf{H}_k^{(i)}$	$3b$	$7b$	/	$4b$
Iteration process	$\mathbf{D}_k^{(i)}$	$2a^2b + 2ab^2 - 2ab - b^2$	$2a^2b + 2ab^2$		
	$\tilde{\mathbf{R}}_k^{(i)}$	b^2	/		
	$\mathbf{P}_{k k-1}^{zz(i+1)}$	$a^2b + ab^2 - ab$	$a^2b + ab^2$		
	$\mathbf{C}_k^{(i+1)1}$	$1.5b^3 - 1.5b^2$	$1.5b^3 + 1.5b^2$		
	$\Delta\mathbf{R}_k^{(i+1)}$	$2b^3 - b^2$	$2b^3$		
	$\mathbf{B}_k^{(i+1)}$	$a^2b + ab^2 - ab + b$	$a^2b + ab^2 + b^2$	/	/
	$\hat{\mathbf{U}}_{k k}^{(i+1)}$	b^2	/	/	/
	$\hat{\mathbf{R}}_k^{(i+1)}$	/	/	b^2	/
	$\mathbf{K}_k^2$	$2a^2b + 2ab^2 - 3ab + 0.5b^3 - 0.5b^2$	$2a^2b + 2ab^2 + 0.5b^3 + 1.5b^2$	/	/
Posterior	$\hat{\mathbf{x}}_{k k}$	$ab + b$	ab	/	/
	$\mathbf{P}_{k k}$	$a^3 + a^2b - a^2$	$a^3 + a^2b$	/	/

¹ $\mathbf{C}_k^{(i+1)}$ costs b FLOP_{SR}s besides the FLOPs given in the table.

² $\mathbf{K}_k$ costs b FLOP_{SR}s besides the FLOPs given in the table.

TABLE IV
COMPARISON OF THE FLOPS FOR THE VB-AEKF AND THE FVB-AEKF FOR DOA TRACKING

Methods	FLOP _A	FLOP _M	FLOP _D	FLOP _{SR}	FLOP _T
VB-AEKF	21635	26525	1606	120	924
FVB-AEKF	≤ 71132	≤ 82211	1604	132	≤ 516

trigonometric functions, the FLOP_Ts carry out much more computational load than the basic operations FLOP_A, FLOP_M, and FLOP_D. Therefore, compared with the VB-AEKF, the FVB-AEKF for DOA tracking significantly improves computational efficiency. This conclusion will be verified by the results of simulations and an experiment in the following sections.

V. SIMULATIONS

In this section, comprehensive simulations were carried out to verify the superiority of the proposed FVB-AEKF for DOA tracking in terms of robustness and computational efficiency.

A. Simulation Scenario

In the simulations, an underwater target was assumed to perform in CV motion mode with the initial state $\mathbf{x}_0 = [90, 0.02]$, and the process noise covariance matrix was $2 \times 10^{-4} \text{ s}^2$. The total simulation time was set to 5000 s with a 1 s interval. Then, the bearing angles of the target are simulated by using (1).

Furthermore, for the measurement model, the frequency f , amplitude a , and the initial phase ϕ_0 of the target signal

were 170 Hz, 1, and 0, respectively, and the sampling frequency of the received signal f_s was 1000 Hz. A 12-element UCA was used to obtain the measurements. The underwater sound velocity c was assumed to be 1500 m/s.

In order to test the robustness of the FVB-AEKF to the unknown measurement noise caused by the unknown underwater environment, a strong Gaussian white noise was added from 3600 to 4000 s of the measurements. Once the MNCM $\mathbf{R}_k$ during the period without the strong noise was set to $\mathbf{I}_P \sigma_r^2$, $\mathbf{R}_k$ at all steps was expressed as

$$\mathbf{R}_k = \begin{cases} \mathbf{I}_P \sigma_r^2, & 0 \leq k < 3600, 4000 < k \leq 5000 \\ 10\mathbf{I}_P \sigma_r^2, & 3600 \leq k \leq 4000 \end{cases} \quad (26)$$

σ_r^2 was set to 0.25 and then the measurement data were simulated by using (5).

The proposed FVB-AEKF for DOA tracking given by Algorithm 2 was tested by the simulated measurement data. The results of the conventional beamforming (CBF)-based DOA estimation [1], the standard EKF, the SH-AEKF for DOA tracking [7], and the VB-AEKF for DOA tracking [23] given by Algorithm 1 were also given for comparison. The scanning bearing angle of the CBF varies every 1° from 0° to 360° . The state estimate of EKF, VB-AEKF, and FVB-AEKF were initialized to $\hat{\mathbf{x}}_0 = [\hat{\theta}_{\text{CBF}}^0, 0]$, where $\hat{\theta}_{\text{CBF}}^0$ was the bearing angle estimated by using CBF at $k = 0$. $\mathbf{P}_{k|k}$ was initialized to $\mathbf{P}_0 = \text{diag}([5, 0.01])$. The process noise covariance matrix $\mathbf{Q}$ for EKF, SH-AEKF, VB-AEKF, and FVB-AEKF were all set to $2 \times 10^{-4} \text{ s}^2$. The MNCM of EKF was set to $\mathbf{I}_P$. The initial estimate of the MNCM and the forgetting factor

Algorithm 2 One Time Step of the Proposed FVB-AEKF Based DOA Tracking Using UCA

Inputs: $\hat{\mathbf{x}}_{k-1|k-1}$, $\mathbf{P}_{k-1|k-1}$, $\mathbf{z}_k$, $\hat{\mathbf{u}}_{k-1|k-1}$, $\hat{\mathbf{U}}_{k-1|k-1}$, $\mathbf{Q}_{k-1}$, ρ , N .

Time update

1. $\hat{\mathbf{x}}_{k-1|k-1} = \mathbf{F}_{k-1|k-1} \hat{\mathbf{x}}_{k-1|k-1}$.
2. $\mathbf{P}_{k|k-1} = \mathbf{F}_{k|k-1} \mathbf{P}_{k-1|k-1} \mathbf{F}_{k|k-1}^T + \mathbf{G}_k \mathbf{Q}_k \mathbf{G}_k^T$.
3. $\hat{\mathbf{u}}_{k-1|k-1} = \rho(\hat{\mathbf{u}}_{k-1|k-1} - n - 1) + n + 1$, $\hat{\mathbf{U}}_{k-1|k-1} = \rho \hat{\mathbf{U}}_{k-1|k-1}$.

Iteration measurement update

4. Initialization: $\hat{\mathbf{x}}_{k|k}^{(0)} = \hat{\mathbf{x}}_{k|k-1}$, $\hat{\mathbf{R}}_k^{(0)} = \hat{\mathbf{U}}_{k|k-1} / \hat{\mathbf{u}}_{k|k-1}$.
5. Calculate $\hat{\mathbf{s}}_{k|k-1}(k)$ and $\mathbf{h}(\hat{\mathbf{x}}_{k|k-1})$.
6. Calculate $\mathbf{H}_{k|k-1}$.
7. $\mathbf{H}_k^{(0)} = \mathbf{H}_{k|k-1}$.
8. $\tilde{\mathbf{P}}_{k|k-1}^{zz} = (\mathbf{z}_k - \mathbf{h}(\hat{\mathbf{x}}_{k|k-1}))(\mathbf{z}_k - \mathbf{h}(\hat{\mathbf{x}}_{k|k-1}))^T$.

For $i = 0 : N - 1$

Update $q^{(i+1)}(\mathbf{R}_k) = \text{IW}(\mathbf{R}_k; \hat{\mathbf{u}}_{k|k}^{(i+1)}, \hat{\mathbf{U}}_{k|k}^{(i+1)})$

If $\varepsilon = \|\mathbf{H}_k^{(i)} - \mathbf{H}_k^{(i-1)}\|_2 < \delta$

9. $\mathbf{D}_k^{(i)} = \mathbf{D}_k^{(i-1)}$, $\mathbf{D}_k^{(i)}$ and $\mathbf{H}_k^{(i)}$ will not be updated in following iterations.

Else

10. Calculate $\hat{\mathbf{s}}_{k|k}^{(i)}(k)$ and $\mathbf{H}_k^{(i)}$.
11. $\mathbf{D}_k^{(i)} = \mathbf{H}_k^{(i)} \mathbf{P}_{k|k-1} (\mathbf{H}_k^{(i)})^T - \mathbf{H}_{k|k-1} \mathbf{P}_{k|k-1} \mathbf{H}_{k|k-1}^T$.
12. $\tilde{\mathbf{R}}_k^{(i)} = \mathbf{D}_k^{(i)} + \hat{\mathbf{R}}_k^{(i)}$,
13. $\mathbf{P}_{k|k-1}^{zz(i+1)} = \mathbf{H}_{k|k-1} \mathbf{P}_{k|k-1} \mathbf{H}_{k|k-1}^T + \tilde{\mathbf{R}}_k^{(i)}$,
14. $\mathbf{C}_k^{(i+1)} = \tilde{\mathbf{R}}_k^{(i)} (\mathbf{P}_{k|k-1}^{zz(i+1)})^{-1}$,
15. $\Delta \mathbf{R}_k^{(i+1)} = \mathbf{C}_k^{(i+1)} (\tilde{\mathbf{P}}_{k|k-1}^{zz} - \mathbf{P}_{k|k-1}^{zz(i+1)}) (\mathbf{C}_k^{(i+1)})^T$,
16. $\mathbf{B}_k^{(i+1)} = \tilde{\mathbf{R}}_k^{(i)} + \Delta \mathbf{R}_k^{(i+1)}$,
17. $\hat{\mathbf{U}}_{k|k}^{(i+1)} = \mathbf{U}_{k|k-1} + \mathbf{B}_k^{(i+1)}$, $\hat{\mathbf{u}}_{k|k}^{(i+1)} = \hat{\mathbf{u}}_{k|k-1} + 1$,
18. $\hat{\mathbf{R}}_k^{(i+1)} = \hat{\mathbf{U}}_{k|k}^{(i+1)} / \hat{\mathbf{u}}_{k|k}^{(i+1)}$.

End for

19. $\hat{\mathbf{R}}_k = \hat{\mathbf{R}}_k^{(N)}$, $\hat{\mathbf{u}}_{k|k} = \hat{\mathbf{u}}_{k|k}^{(N)}$, $\hat{\mathbf{U}}_{k|k} = \hat{\mathbf{U}}_{k|k}^{(N)}$.

Update $q(\mathbf{x}_k) = \text{N}(\mathbf{x}_k; \mathbf{x}_{k|k}, \mathbf{P}_{k|k})$

20. $\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}_{k|k-1}^T (\mathbf{H}_{k|k-1} \mathbf{P}_{k|k-1} \mathbf{H}_{k|k-1}^T + \hat{\mathbf{R}}_k)^{-1}$,
21. $\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k (\mathbf{z}_k - \mathbf{h}(\hat{\mathbf{x}}_{k|k-1}))$,
22. $\mathbf{P}_{k|k} = \mathbf{P}_{k|k-1} - \mathbf{K}_k \mathbf{H}_{k|k-1} \mathbf{P}_{k|k-1}$.

Outputs: $\hat{\mathbf{x}}_{k|k}$, $\mathbf{P}_{k|k}$, $\hat{\mathbf{u}}_{k|k}$, $\hat{\mathbf{U}}_{k|k}$.

for the SH-AEKF was set to $\mathbf{I}_P$ and 0.9999, respectively. For the VB-AEKF and FVB-AEKF, the dof $\hat{\mathbf{u}}_0$ and the inverse scale matrix $\hat{\mathbf{U}}_0$ of the inverse Wishart distribution were set to 16 and $16\mathbf{I}_P$, respectively, and the forgetting factor ρ was set to 0.9975. The VB iteration number was set to 10.

In order to statistically evaluate the performance of the FVB-AEKF, 500 Monte Carlo simulations were performed on a PC configured with an Intel Core i7-9700F CPU (at 3.0 GHz) and a RAM of 16 GB. The root mean square error (RMSE)

for the DOA tracking results at time step k was defined as

$$\text{RMSE}_\theta = \sqrt{\frac{1}{N} \sum_{n=1}^N (\hat{\theta}_{k,n} - \theta_k)^2} \quad (27)$$

where N is the total number of the Monte Carlo trials, $\hat{\theta}_{k,n}$ is the bearing angle estimate of the target in the n th trail, θ_k is the true bearing angle of the target.

To evaluate the precision of the FVB-AEKF in total time, the average RMSE for the DOA tracking results in total time was defined as:

$$\overline{\text{RMSE}}_\theta = \sqrt{\frac{1}{KN} \sum_{k=1}^K \sum_{n=1}^N (\hat{\theta}_{k,n} - \theta_k)^2} \quad (28)$$

where K is the total number of the time steps. A less RMSE represents a higher precision of the DOA tracking.

B. Verification of VB-AEKF and FVB-AEKF

1) Bearing Angle Estimate Results: The bearing angle tracking results of the FVB-AEKF for DOA tracking are shown in Fig. 2. The results of the CBF, the EKF, the SH-AEKF, and the VB-AEKF are also shown in Fig. 2 for comparison.

Fig. 2 shows that the performance of the CBF was acceptable when the measurement noise was weak and stationary. However, the CBF was invalid during the strong noise from 3600 to 4000 s due to its excessive reliance on the present measurement. The trajectory obtained by the DOA tracking methods EKF, SH-AEKF, VB-AEKF, and FVB-AEKF fluctuated in the beginning and stabilized after about 1000 s, which means these algorithms converged after 1000 s. The time for stabilization cost by these DOA tracking methods is acceptable in the application. The reason is that some errors existed in the initial state estimate $\hat{\mathbf{x}}_0$ and the mean square error matrix $\mathbf{P}_0$, which affect the tracking performance at the beginning. After these methods were performed for a period of time, the error was gradually eliminated by the correction of the new measurements. The precision of the EKF was higher than that of the CBF-based DOA estimation after the algorithm converged. Furthermore, since EKF considers the kinematic model of the underwater target, the EKF provided valid tracking even from 3600 to 4000 s. The SH-AEKF is less affected by the strong noise because it provides more accurate Kalman filter gains due to the estimation of the MNCM, which can significantly improve the tracking performance. The tracking path of the FVB-AEKF was largely consistent with the VB-AEKF since the FVB-AEKF is another mathematical expression of the VB-AEKF. Moreover, the VB-AEKF and the FVB-AEKF provided the most high-precision tracking. The reason is that the VB-AEKF and the FVB-AEKF give the prior distribution of MNCM, and uses the VB iteration method to jointly estimate the MNCM and the target state, which improve the accuracy of MNCM estimation.

2) RMSE Analysis: In order to statistically evaluate the performance of the proposed FVB-AEKF, Monte Carlo simulations were carried out. The RMSEs of the bearing angle estimates and the run time of the methods are given and analyzed. Fig. 3(a) shows the RMSEs of the bearing angle

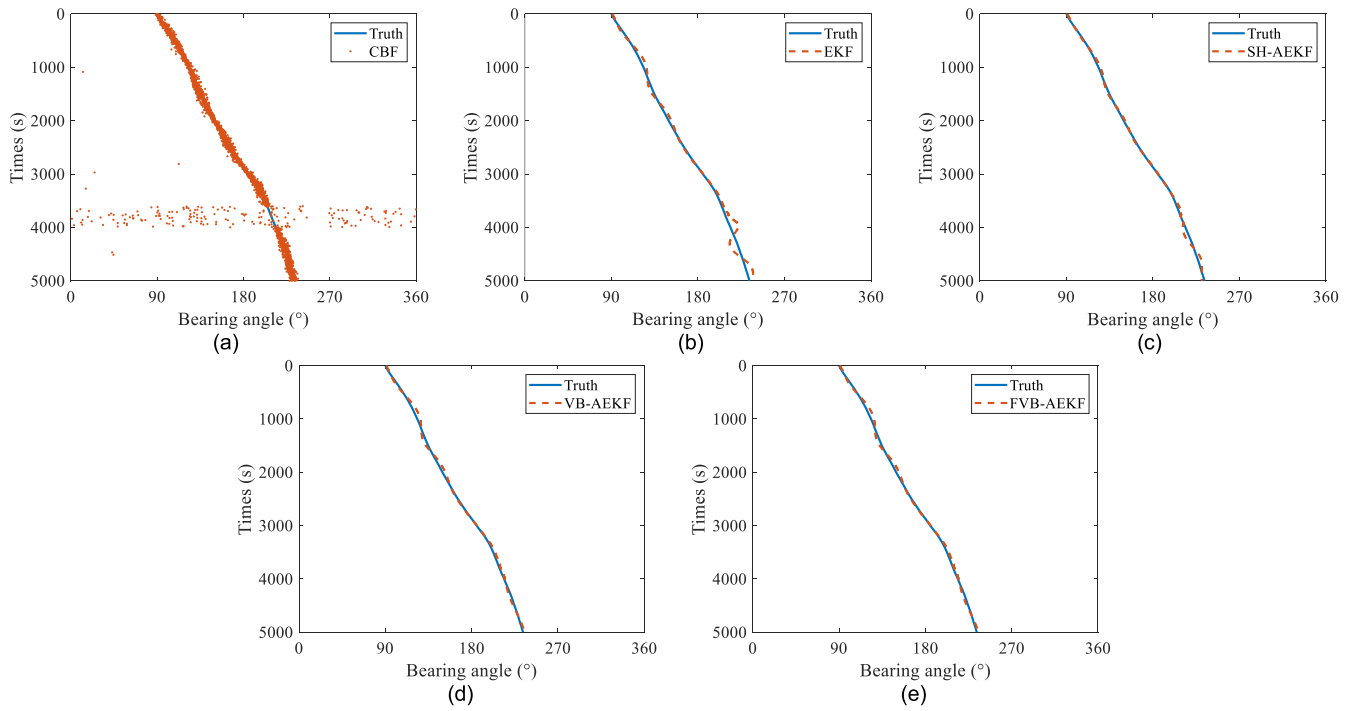


Fig. 2. Bearing angle tracking results of (a) CBF, (b) EKF, (c) SH-AEKF, (d) VB-AEKF, and (e) FVB-AEKF.

TABLE V

AVERAGE RMSEs AND MEANS OF THE RUN TIME IN ONE TIME STEP OF THE CBF, EKF, VB-AEKF, AND FVB-AEKF

	CBF	EKF	SH-AEKF	VB-AEKF	FVB-AEKF
$RMSE_{\theta}$ (°)	27.55	1.52	1.24	1.09	1.10
Mean of run time (ms)	0.91	0.05	0.06	0.85	0.30

estimates of the FVB-AEKF at all time steps, and Fig. 3(b) gives the partially magnified results for details. Table V shows the average RMSEs and the mean of the run time in one time step. The results of the CBF, the EKF, SH-AEKF, and VB-AEKF are also given for comparison.

Fig. 3 and the average RMSEs results in Table V show that similar to the conclusion in the previous subsection, the DOA tracking methods EKF, VB-AEKF, and FVB-AEKF converged after a period of time. The tracking precision of SH-AEKF is superior to that of EKF. The VB-AEKF and the FVB-AEKF provide the most high-precision tracking even during the existence of the strong noise, while the CBF was invalid during this time. The superior performance of the proposed VB-AEKF and the FVB-AEKF was verified again.

Table V also shows that the CBF cost the longest time because it needs to scan all bearing angles in space. The EKF cost the least time because of its simple computation. Since the SH-AEKF cost time to estimate the MNCM, the runtime of it was longer than EKF. The VB-AEKF cost more time than EKF and SH-AEKF because of its iteration process. The run time of the FVB-EKF decreased by 65% compared to the VB-AEKF. The reason is that the FVB-AEKF significantly reduces the computations of the FLOPs of trigonometric functions with high computational complexity. What is more, the tracking

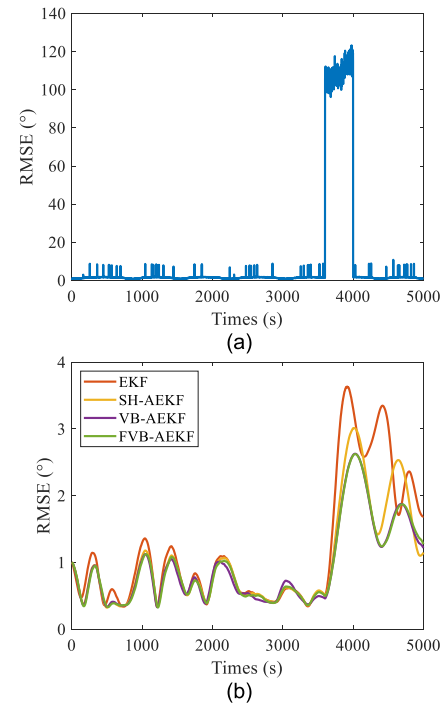


Fig. 3. RMSEs of (a) CBF and (b) EKF, SH-AEKF, VB-AEKF, and FVB-AEKF at all time steps.

performance of the VB-AEKF and the FVB-AEKF were almost the same. The superiority of FVB-AEKF in terms of computational efficiency and precision was then verified. The conclusion above is consistent with the theoretical analysis in Section IV. In addition, it should be noticed that, although the FVB-EKF cost more time than EKF and SH-AEKF, FVB-EKF provided more accurate tracking than EKF and SH-AEKF.

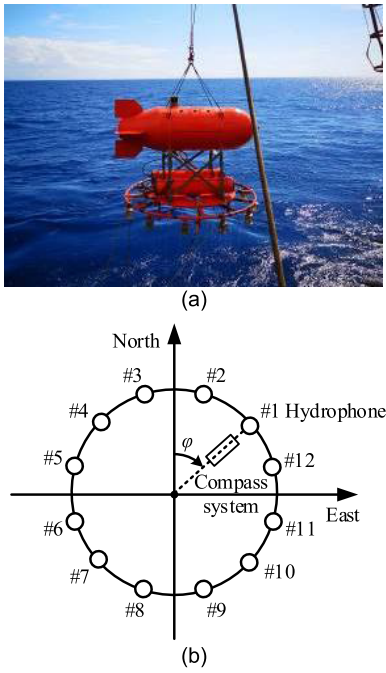


Fig. 4. Configuration of the underwater buoy system. (a) Photograph of the underwater buoy system. (b) Configuration of the UCA and the compass system.

VI. EXPERIMENTAL RESULTS

A. Experimental Setup and Description

The data from an acoustic experiment in the South China Sea in July 2021 was used to test the FVB-AEKF. An acoustic signal emission system fixed on a ship that consists of a signal generator, a power amplifier, and an acoustic emission transducer was considered as a target. The signal generator and the power amplifier were deployed on the board and the acoustic emission transducer was placed 3 m under the sea surface. The electric signal output by the signal generator was amplified by the power amplifier and then transduced to the acoustic signal by the acoustic emission transducer. The ship equipped with the acoustic signal emission system kept moving when an acoustic signal at 170 Hz was emitted continuously for 7 500 s with a sound source level of 145 dB.

An underwater buoy system shown in Fig. 4(a) was placed at 20 m under the sea surface before the experiment. The distance from it to the acoustic signal emission system satisfied the far-field assumption. A uniform circular hydrophone array with a radius of 1 m with 12 elements and a digital acquisition system was fixed on the underwater buoy. Elements 1 to 12 were placed counterclockwise around the periphery of the circle. Considering the underwater buoy system rotated with the waves, a compass system was fixed on it as shown in Fig. 4(b) to measure the bearing of hydrophone 1 to offset the rotation angle. In Fig. 4(b), the rectangle represents the compass system, and φ (measured clockwise with respect to the north) represents the bearing of element 1. During emitting the acoustic signal, the digital acquisition system sampled the output of the circular hydrophone array at a frequency of 8 192 Hz and the compass system measured the bearing angles of hydrophone 1 every second. The sound

TABLE VI
ABEES OF THE CBF, EKF, SH-AEK, VB-AEK, AND FVB-AEK

Data	ABEE (°)				
	CBF	EKF	SH-AEK	VB-AEK	FVB-AEK
Raw data	13.2	8.2	7.1	6.3	6.3
With noise	19.9	12.2	7.3	6.5	6.5

velocity was measured to be constant at 1544 m/s at the depth from 0 to 20 m with a conductivity, temperature, depth (CTD) profiler before the experiment.

In order to provide the true bearing angle of the target relative to the circular hydrophone array for reference, the acoustic signal emission system and the underwater buoy system were equipped with Global Positioning Systems (GPS). The GPSs recorded the locations of the acoustic signal emission system and the underwater buoy system every second. Considering the bearing angle of the target is measured clockwise with respect to the north, the bearing angle trajectory is calculated by the GPS data.

B. Comparison of CBF, EKF, VB-AEK, and FVB-AEK for DOA Tracking

The VB-AEK and the FVB-AEK were tested using the experimental data of all 7 500 s. The CBF-based DOA estimation, the EKF, SH-AEK, VB-AEK, and FVB-AEK were also carried out for comparison. The parameters are set to be the same as those in the simulation. In order to numerically compare the precision of the bearing angle estimation, the bearing angle estimation error (BEE) and the averaged BEE (ABEE) were defined as the performance metrics, which are given as follows:

$$\text{BEE}(k) = \left| \hat{\theta}_k - \theta_k^{\text{GPS}} \right| \quad (29)$$

$$\text{ABEE}(k) = \sqrt{\frac{1}{K} \sum_{k=1}^K \left(\hat{\theta}_k - \theta_k^{\text{GPS}} \right)^2} \quad (30)$$

where $\hat{\theta}_k$ is the bearing angle estimate at time k and θ_k^{GPS} is the true bearing angle obtained by the GPS data.

The trajectories obtained by the CBF, EKF, SH-AEK, VB-AEK, and FVB-AEK are shown in Fig. 5(a). The BEEs and the ABEEs are shown in Fig. 5(b) and Table VI, respectively. Specifically, considering the complexity of the actual underwater environment, the CBF obtained bearing angles every 10 s by using ten measurements as snapshots for a higher precision [7]. However, this higher precision is obtained at the expense of the real-time tracking capability. In order to further test the robustness of the proposed FVB-AEK to unknown measurement noise, a period of Gaussian noise with the covariance of $2\mathbf{I}_P$ was added to the raw measurement data from 2600 to 3000 s, and then the methods were tested again. The results are shown in Fig. 5(c) and (d) and Table VI. The black dotted lines in Fig. 5(c) and (d) denote the beginning and ending times of the added noise.

Fig. 5(a) and (b) show that the trajectories obtained by all methods using raw experimental data fluctuated around the true trajectory due to the effect of the underwater environmental noise. Fig. 5(a) and (b) and Table VI shows

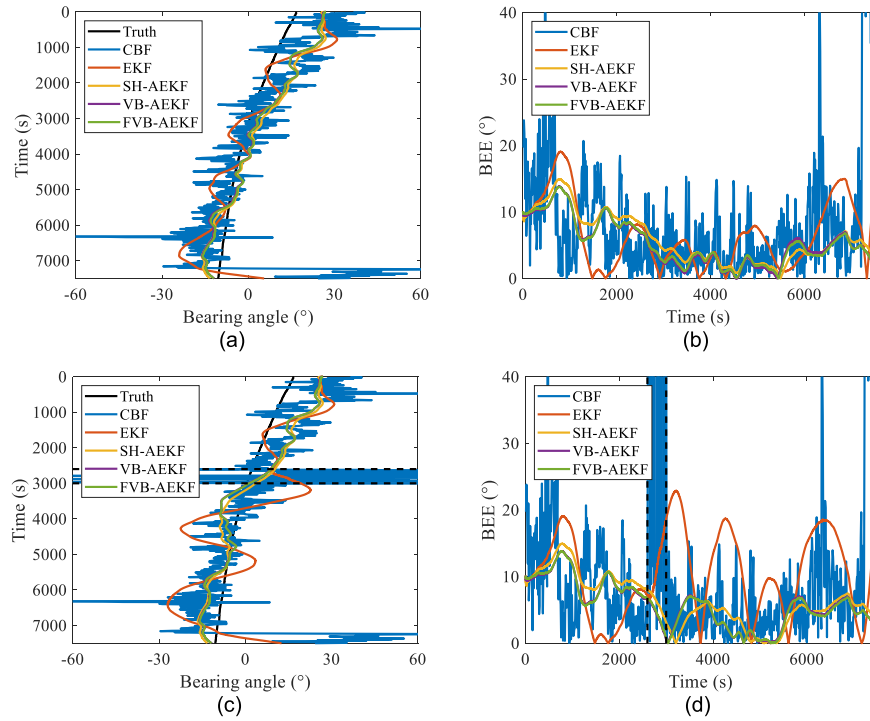


Fig. 5. Experimental results of CBF-based DOA estimation, EKF, SH-AEKF, VB-AEKF, and FVB-AEKF. (a) Trajectories obtained using raw data. (b) BEEs obtained using raw data. (c) Trajectories obtained using the data with added noise. (d) BEEs obtained using the data with added noise.

that the VB-AEKF and the FVB-AEKF provided the most accurate tracking of all the methods. The reason is that the VB-AEKF and the FVB-AEKF not only predict the bearing angle of the target using the kinematic model of the underwater target but also accurately estimate the MNCMs. The tracking precision of SH-AEKF is lower than VB-AEKF and FVB-AEKF because the VB iteration process provided more accurate MNCMs than the Sage-Husa algorithm. The trajectory obtained by the EKF fluctuated more than SH-AEKF, VB-AEKF, and FVB-AEKF due to the mismatching between the known and stationary MNCM assumption and the actual unknown and unstable MNCM. The CBF-based DOA estimation provided the trajectory with the largest error because of ignoring the kinematic information of the underwater target.

The trajectories obtained using the data with added noise given in Fig. 5(c) and (d) show a similar conclusion to the above. Furthermore, the VB-AEKF and FVB-AEKF still provided robust tracking even after the added noise presented, while the performance of the EKF and the CBF-based DOA estimation obviously degraded. The performance of SH-AEKF is also slightly affected by the added noise, but the precision of SH-AEKF is lower than VB-AEKF and FVB-AEKF. The ABEEs obtained using the data with added noise given in Table VI show the superiority of the VB-AEKF and FVB-AEKF in the precision again.

The run time in the one time step of the CBF-based DOA estimation, the EKF, the VB-AEKF, and the FVB-AEKF are shown in Table VII. The run time of the FVB-EKF decreased by half compared to the VB-AEKF in both the process of the raw data and that of the data with added noise. This

TABLE VII
RUN TIMES IN ONE TIME STEP OF CBF, EKF, SH-AEKF, VB-AEKF, AND FVB-AEKF

Data	Run time (ms)				
	CBF	EKF	SH-AEKF	VB-AEKF	FVB-AEKF
Raw data	1.12	0.11	0.13	0.70	0.34
With noise	1.08	0.11	0.14	0.75	0.35

shows once again that the FVB-EKF can provide higher computational efficiency than VB-EKF while the performance loss introduced by replacing the VB-AEKF with the FVB-EKF can be ignored. The EKF and SH-AEKF cost less time than the FVB-EKF, but provide DOA trajectories with larger errors. Thus, FVB-AEKF shows superiority in terms of robustness and precision.

VII. CONCLUSION

The robust underwater DOA tracking with a nonlinear measurement model in the scenario of unknown measurement noise is studied in this article. First, for the sake of accurate tracking and minimum underwater acoustic resource utilization, the kinematic model of the underwater target by bearing angle and the nonlinear measurement model based on the received signal of a UCA are designed. Then, based on the determined model, the algorithm of VB-AEKF for robust underwater DOA tracking with unknown measurement noise is given. Finally, by deriving the equivalent process with less computations of VB-AEKF, the FVB-AEKF for DOA tracking with nonlinear and high-dimensional measurement is proposed. The FVB-AEKF improves computational efficiency

while tracking accuracy is ensured. The simulations and the experiment verified the superior tracking performance and the high efficiency of the FVB-AEKF in underwater DOA tracking scenarios. It is believed that the proposed FVB-AEKF has significant potential in real-time underwater DOA tracking missions.

APPENDIX

The estimate of the signal $\hat{s}_{k|k}^{(i)}(k)$ is obtained by using the ML approach as [5]

$$\hat{s}_{k|k}^{(i)}(k) = \left(A^H \left(\theta_{k|k}^{(i)} \right) A \left(\theta_{k|k}^{(i)} \right) \right)^{-1} A^H \left(\theta_{k|k}^{(i)} \right) \text{Hilbert}(z(k)) \quad (31)$$

where $A(\theta_{k|k}^{(i)})$ is the array manifold vector given by (6), $(\cdot)^H$ and $\text{Hilbert}(\cdot)$ denote the conjugate transposition and the Hilbert transform, respectively.

The estimate of the measurement function $\mathbf{h}(\hat{\mathbf{x}}_{k|k}^{(i)})$ is given by

$$\mathbf{h}(\hat{\mathbf{x}}_{k|k}^{(i)}) = \text{real}(A(\theta_{k|k}^{(i)})\hat{s}_{k|k}^{(i)}(k))$$

where $\text{real}(\cdot)$ denotes the real part.

The Jacobian matrix of the measurement function, $\mathbf{H}_k^{(i)}$, is given by [7]

$$\mathbf{H}_k^{(i)} = \frac{\partial \mathbf{h}(\hat{\mathbf{x}}_{k|k}^{(i)})}{\partial \hat{\mathbf{x}}_{k|k}^{(i)}} = \left[\frac{\partial \mathbf{h}(\hat{\mathbf{x}}_{k|k}^{(i)})}{\partial \hat{\theta}_{k|k}^{(i)}}, \mathbf{0} \right] = \left[h'_1, h'_2, \dots, h'_P \right]^T. \quad (32)$$

The element of the matrix h'_p ($p = 1, 2, \dots, P$) is expressed as

$$h'_p = -2\pi f r / c \sin(\hat{\theta}_{k|k}^{(i)} - 2\pi p / P) \text{real}(j A_p(\hat{\theta}_{k|k}^{(i)}) \hat{s}_{k|k}^{(i)}). \quad (33)$$

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